

Linear Regression

Practical walkthrough: Exam score vs study time

Includes: dataset tables • fitted equations • evaluation metrics • diagnostic plots

What linear regression does

A simple, interpretable model for predicting a numeric outcome

Goal

Learn a straight-line relationship between inputs (features) and an output (target) by minimizing squared prediction error.

Simple (1 feature):

$$\hat{y} = \beta_0 + \beta_1 \cdot x$$

Multiple (k features): $\hat{y} = \beta_0 + \beta_1 \cdot x_1 + \dots + \beta_k \cdot x_k$

Interpretation (what coefficients mean)

- β_1 is the average change in the prediction for a +1 change in x (holding other features constant).
- β_0 is the baseline prediction when all x are 0 (may be outside the realistic range).
- The model is fit by minimizing $\sum (y - \hat{y})^2$ across the dataset (ordinary least squares).

We will model

Exam score (0–100) from study time (hours). Then we add sleep + attendance as extra predictors.

Inputs (features)

- study_hours
- sleep_hours
- attendance_rate



Linear model

$$\hat{y} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_3$$



Output (target)

exam_score

Dataset

180 students (synthetic) • score 0–100 • study hours 0–30

Sample rows

Sample rows (first 10 students)

exam_score	study_hours	sleep_hours	attendance_rate
47.4	11.0	7.1	0.83
39.7	7.1	6.3	0.88
54.3	6.6	8.2	0.93
56.2	6.6	8.6	0.94
65.8	20.8	6.6	0.91
55.2	13.1	6.6	0.79
50.3	5.4	7.3	0.86
64.1	11.4	7.0	1.0
48.5	9.3	7.9	0.94
33.0	1.2	6.5	0.82

Why these variables?

- study_hours: time spent preparing (main feature).
- sleep_hours: rest affects performance (often overlooked confounder).
- attendance_rate: proxy for engagement / exposure to material.
- exam_score: target outcome.

Summary statistics

Summary statistics

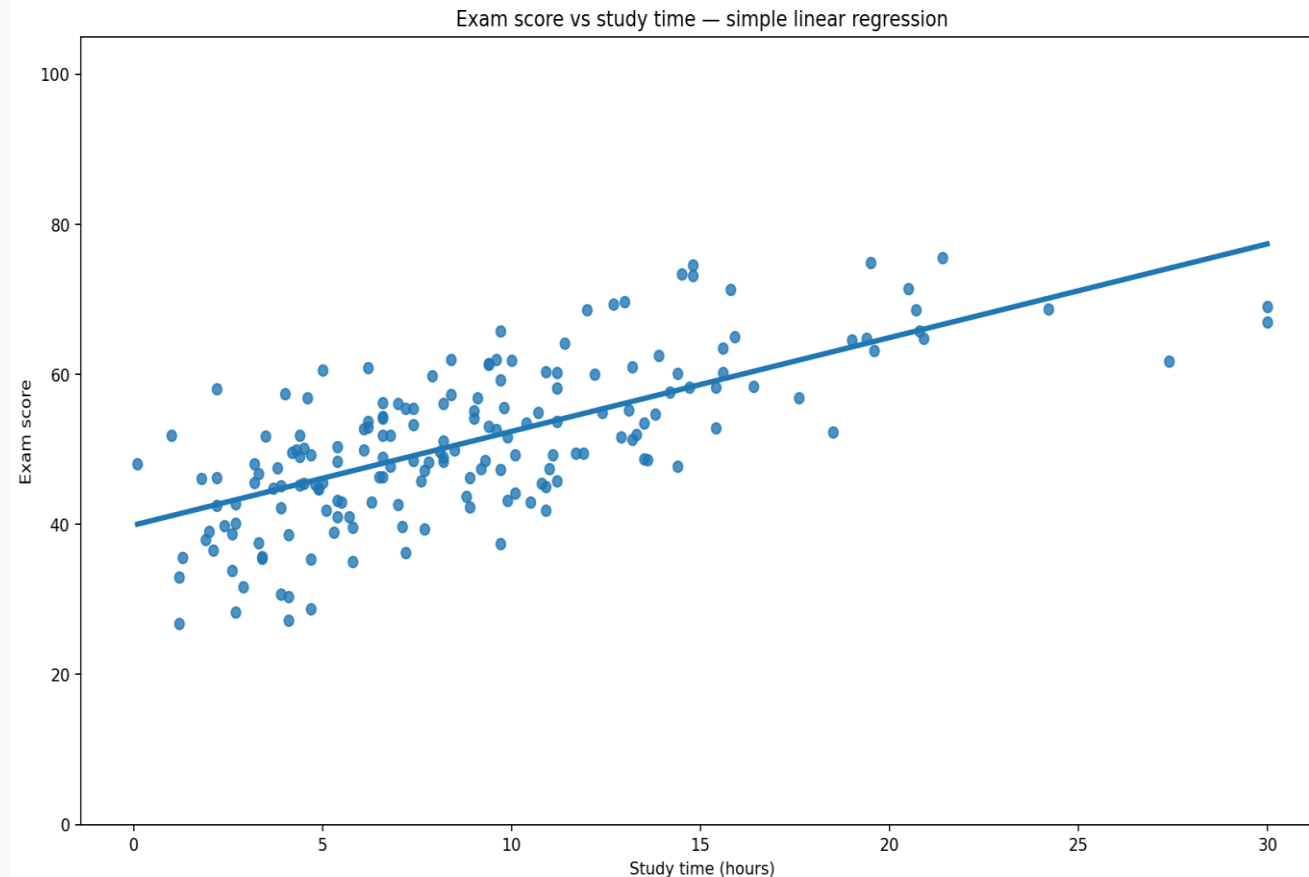
stat	exam_score	study_hours	sleep_hours	attendance_rate
mean	50.91	8.99	7.06	0.86
std	10.35	5.59	1.02	0.08
min	26.8	0.1	4.5	0.64
25%	44.78	4.7	6.5	0.81
50%	49.9	8.15	7.0	0.86
75%	57.7	11.75	7.82	0.91
max	75.5	30.0	9.4	1.0

Note: Data is synthetic (generated) to demonstrate workflow—replace with your real class data to apply directly.

Exploratory view

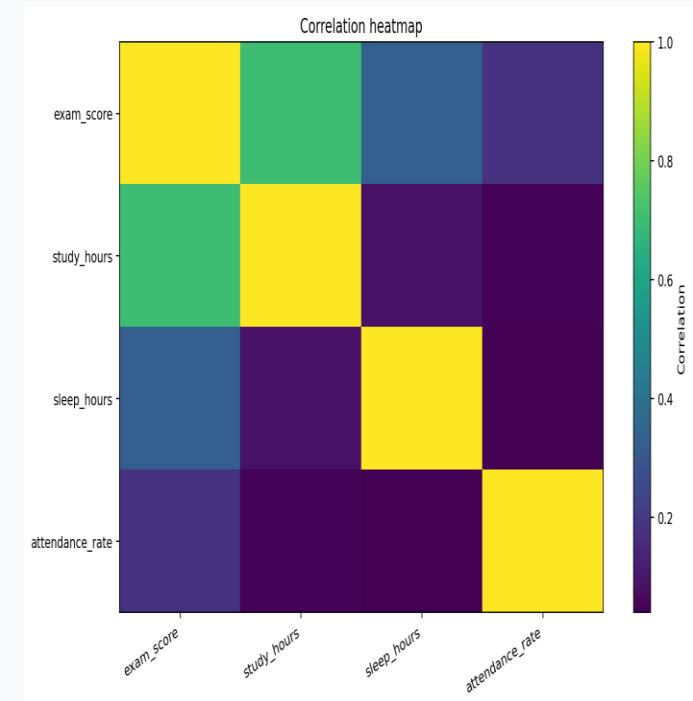
Before fitting: check trend, spread, and correlations

Exam score vs study time



What to look for: upward trend, outliers, and whether a straight line seems reasonable.

Correlations



- Correlation $\neq$ causation.
- Study time correlates with score, but sleep and attendance also matter.
- If residuals show patterns later, it may indicate non-linearity (e.g., diminishing returns).

Simple linear regression

Model: exam_score ~ study_hours

Fitted equation

$$\hat{y} = 39.890 + 1.249 \cdot \text{study_hours}$$

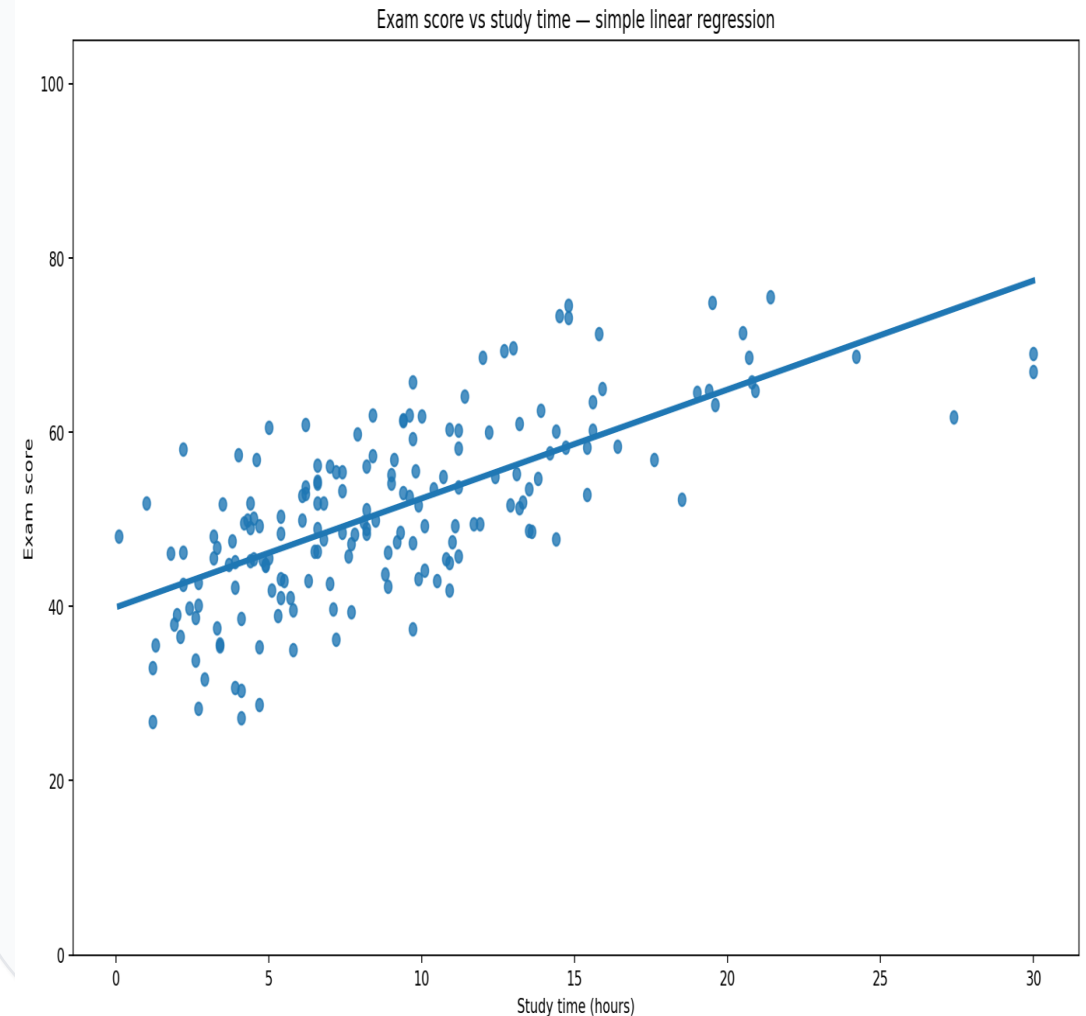
How to interpret β_1 (slope)

- About +1.25 points per additional hour studied (on average).
- A +5 hour increase corresponds to ~+6.2 points (holding other factors constant — though they are not in this model).
- Intercept (~39.9) is the predicted score at 0 study hours (may not be realistic for everyone).

What the model does NOT capture

- Diminishing returns (studying more helps, but the gain may taper).
- Other important predictors: sleep, attendance, prior skill, test anxiety, etc.

Fit on the data



How good is the simple model?

Evaluate on a held-out test set (20% of the data)

Metrics (test set)

Test-set metrics — simple regression

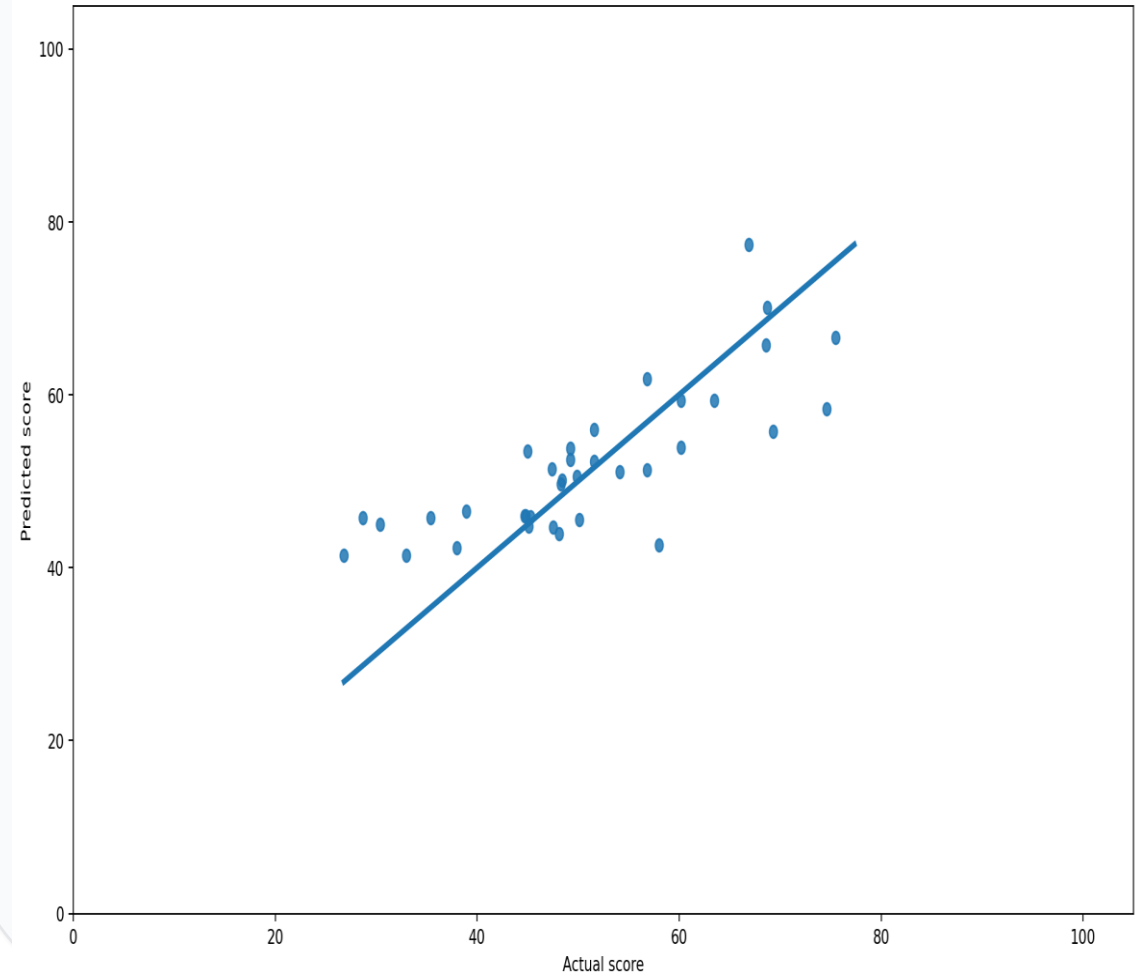
metric	value
R^2	0.601
MAE (pts)	5.963
RMSE (pts)	7.765

What these mean

- R^2 : fraction of score variance explained by study time.
- MAE: typical absolute error (in score points).
- RMSE: like MAE but penalizes large errors more.
- Always evaluate on unseen data to avoid being fooled by overfitting.

Predicted vs actual (test set)

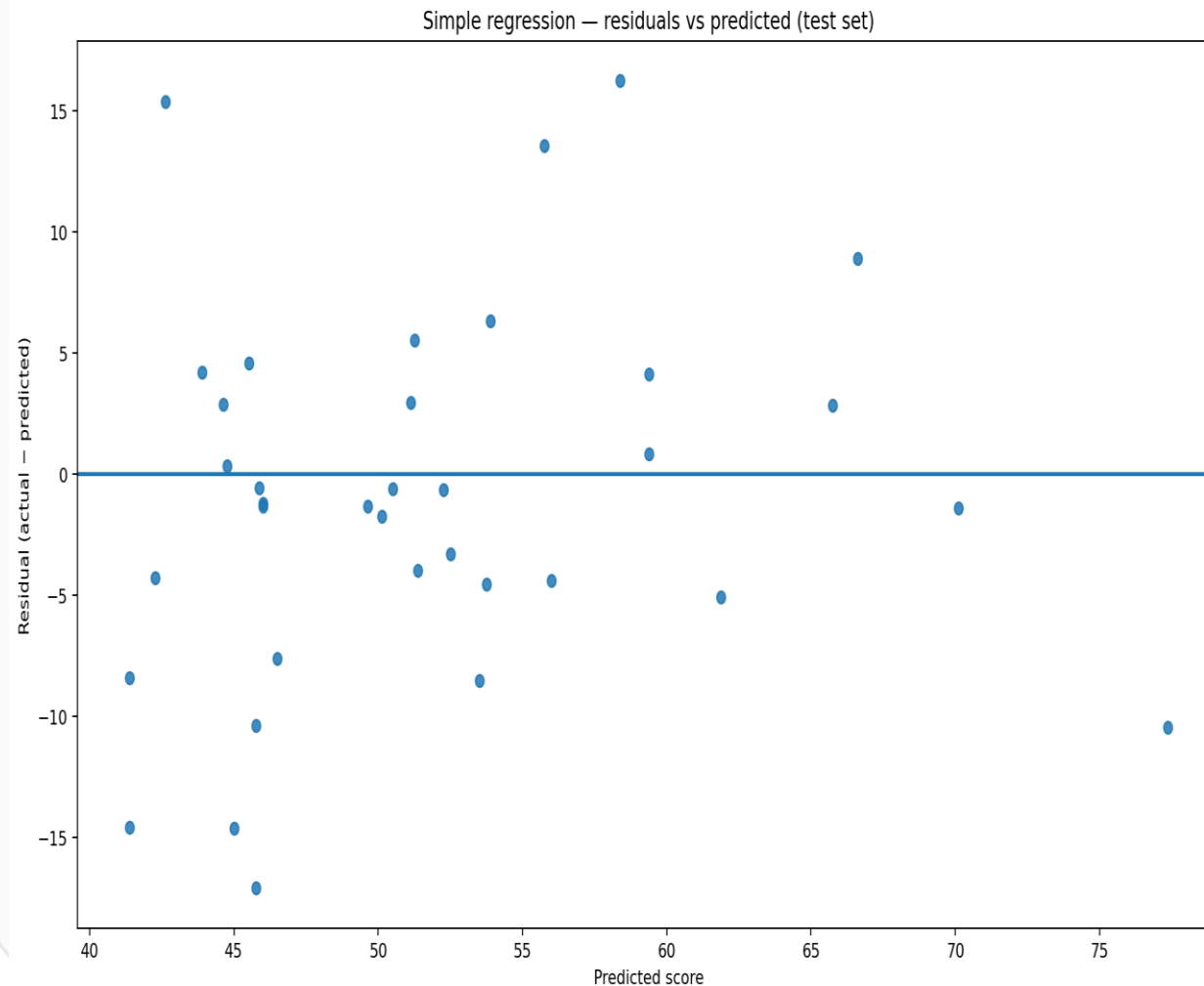
Simple regression — predicted vs actual (test set)



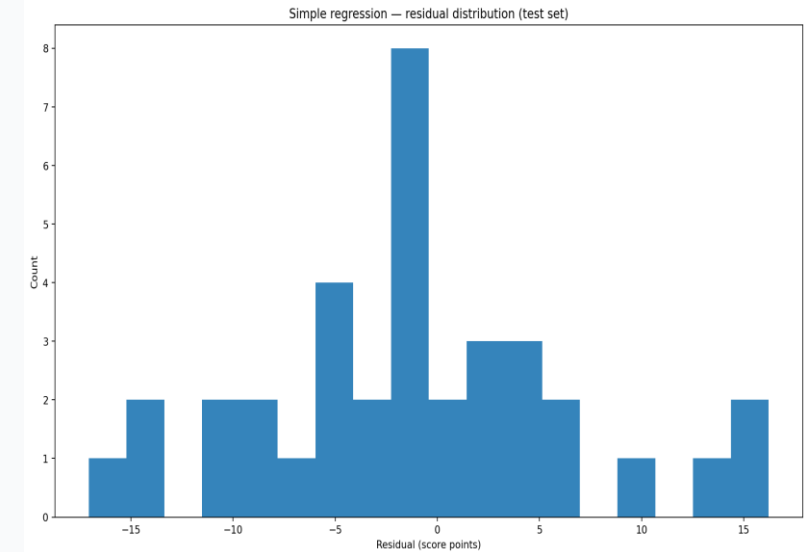
Diagnostics

Residuals help you spot missing structure (non-linearity, heteroscedasticity, outliers)

Residuals vs predicted



Residual distribution



How to read these

- Ideally, residuals look like random noise around 0.
- Curves or patterns suggest non-linearity (e.g., diminishing returns).
- Fan-shaped spread suggests changing variance (heteroscedasticity).
- A skewed histogram suggests systematic bias for some students.

Multiple regression

Add sleep + attendance to explain more of the score

Model

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{study} + \beta_2 \cdot \text{sleep} + \beta_3 \cdot \text{attendance}$$

Fitted (this dataset):

$$\hat{y} = 4.166 + 1.155 \cdot \text{study} + 2.832 \cdot \text{sleep} + 19.528 \cdot \text{attendance}$$

Coefficients

Fitted coefficients — multiple regression

term	coefficient
intercept	4.166
study_hours	1.155
sleep_hours	2.832
attendance_rate	19.528

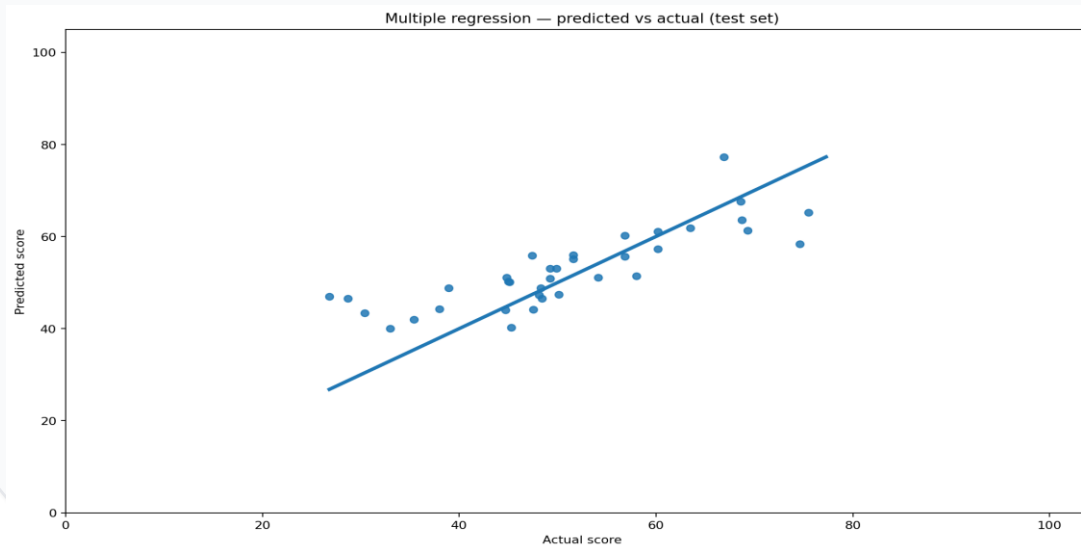
- Study: +1 hour → ~+1.16 points (holding sleep & attendance fixed).
- Sleep: +1 hour/night average → ~+2.83 points.
- Attendance: +0.10 (10%) → ~+1.95 points (because 19.53×0.10).

Performance (test set)

Test-set metrics — multiple regression

metric	value
R ²	0.625
MAE (pts)	5.776
RMSE (pts)	7.53

Predicted vs actual



Simple vs multiple regression

Adding relevant features usually improves fit and reduces error

Model comparison (test set)

Model comparison (test set)

model	R^2	MAE (pts)	RMSE (pts)
Simple: <code>score~study_hours</code>	0.601	5.963	7.765
Multiple: <code>+sleep+attendance</code>	0.625	5.776	7.53

What changed and why

- R^2 improved (explains more score variation) because sleep and attendance carry extra information.
- Errors (MAE/RMSE) decreased modestly: the model misses fewer points on average.
- Interpretation becomes “holding other factors constant,” which is often closer to how we reason about performance.
- If you add too many weak/noisy features, you can overfit—use cross-validation and regularization when needed.

Using this on your real class data

A practical checklist

Workflow

- Collect data: study hours + exam scores (and any relevant factors).
- Plot score vs hours; check for outliers and non-linear trends.
- Split train/test or use cross-validation.
- Fit a baseline model (simple).
- Add a few meaningful predictors (multiple).
- Check residuals; if curved, try transforms or polynomial features.
- Report metrics (R^2 /MAE/RMSE) + coefficient interpretations.

Common pitfalls

- Range restriction: if everyone studies 8–12 hours, slope estimates become noisy.
- Confounding: omitted variables can distort the “study → score” relationship.
- Leakage: don’t compute features using test-set information

One-slide recap

Linear regression =

a best-fit line/plane that predicts a number
fit by minimizing squared error
and interpreted via coefficients

Simple example:

$$\hat{y} = 39.890 + 1.249 \cdot \text{study_hours}$$

Always validate on unseen data and check residuals.

Next steps if needed: try polynomial regression or regularization (Ridge/Lasso).